

Information Propagation Methods for Reliability, Reachability and Flow Analysis in Directed Acyclic Process Networks

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Who it is for, and the problem

The engineer responsible for a process system, such as a water treatment works and its distribution network, asks many questions of it.

This thesis addresses three:

- Will supply reach each point? (*reliability*)
- How much can it deliver, and what limits it? (*capacity*)
- How long to restore it, and what is critical? (*schedule*)

Each has its own method and software, so the **topology is entered three times** and kept consistent by hand.

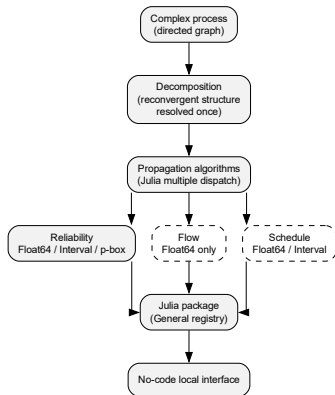
What the engineer actually knows

- a failure probability from three recorded failures
- a nameplate capacity with a tolerance
- a duration given as a range, not a number

The exact methods in common use take one number per input, so the uncertainty is **collapsed** before the analysis runs.

And the system model is often sensitive: it should **stay on the engineer's machine**.

One model, three interpretations

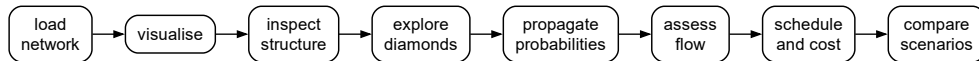


A system is a **DAG**: components are nodes, dependencies are edges. One **graph object** is read by three toolkits; only the meaning of the value on a component changes, a failure probability, a throughput limit or a duration. A further analysis would read the same object.

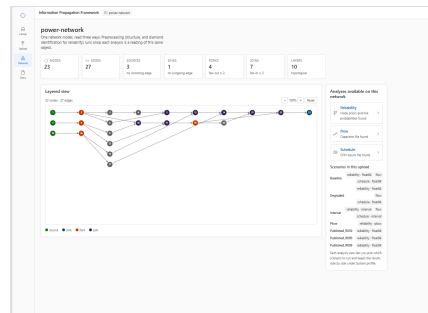
Research questions

- 1 Can one DAG model serve all three analyses, topology entered once?
- 2 Can reconvergence be identified once and reused?
- 3 Exact reliability under intervals and p-boxes: what is guaranteed, at what cost?
- 4 Which critical-path quantities stay exact under interval durations?
- 5 Can it reach an engineer who does not program, with the model kept on their machine?

What the engineer does with it

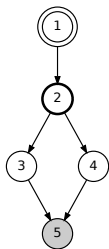


- Uploads the files they already have: an edge list and one JSON file per analysis. No code.
- Every result keeps the form its inputs were given in, an interval stays an interval, and names its method: *exact*, *sound* or *conservative enclosure*.
- Runs on their own machine: a browser interface over a loopback server.
- The same methods are a registered Julia package, `InformationPropagationAnalysis.jl`, for scripted use.



the network page after upload

Will supply reach each point? Redundancy is what makes it hard



every component 0.9

Redundant routes split and rejoin, sharing what is upstream.
Treating them as independent **overstates** reliability: $b(5) = 0.749$ against the exact 0.675.

Conditioning on the fork's reachability repairs it exactly:

$$b(5) = 0.9 \left[0.81 \times (1 - (1 - 0.729)^2) + 0.19 \times 0 \right] = 0.675462$$

Two sub-problems replace 2^9 states of the uncertain components.

The **Network Decomposition Module** finds every such *diamond* once, with its **conditioning set**, proved complete, deterministic and terminating.

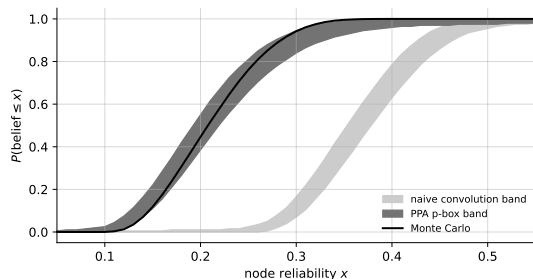
How sure can the engineer be? Reliability with uncertain inputs

One pass over the network, conditioning at each join. The answer comes back in the form the inputs were given:

- **numbers:** the exact reliability of every node;
- **intervals:** the exact range, because reliability is monotone in each input;
- **p-boxes:** sound bounds on its distribution.

Agrees with a decision-diagram oracle to 1.1×10^{-16} on 129 graphs.

Cost: exponential in the largest conditioning set, the same width a decision diagram pays. The gain is the uncertainty carried through, not speed.

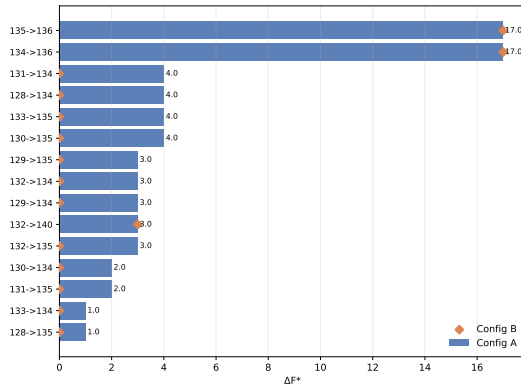


the p-box band brackets the Monte Carlo distribution

How much can it deliver, and what limits it?

Capacity (point-valued)

- Throughput, every minimum cut, saturated edges, single points of failure, how far an edge can degrade or must be upgraded.
- All minimum cuts from one solve: the closed subsets of the free zone (Picard and Queyranne), at most $2^{|\mathcal{F}|}$.
- Two designs deliver the same 37 units; one has 4 minimum cuts, the other 1. Where the network binds differs, and so does what to reinforce.

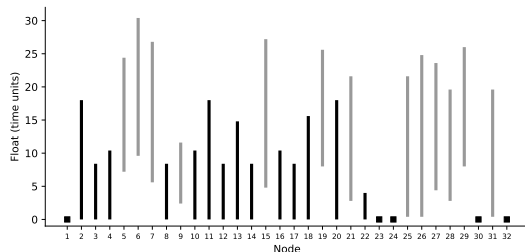


loss in throughput when one edge fails: design A (bars) has 15 edges that matter, design B (markers) has 3

How long to restore it, and what is critical?

Schedule

- Critical path generalised to a family of operator pairs; with interval durations, the completion time is exact from two runs.
- Which activities are critical *whatever* the durations turn out to be (*necessarily*), or only for some (*possibly*)? In general this means trying every duration at its low or high end: 2^k runs, and NP-hard. The **domination split** varies only the durations that can route around each activity.
- 30-activity PSPLIB benchmark: 50,524 runs instead of 2^{30} (≈ 1.1 billion). It found 3 activities critical whatever happens, which the fast conservative method could not confirm, and cleared 7 that it had flagged.



exact float range of each activity, durations $\pm 20\%$:
points at zero are critical in every case

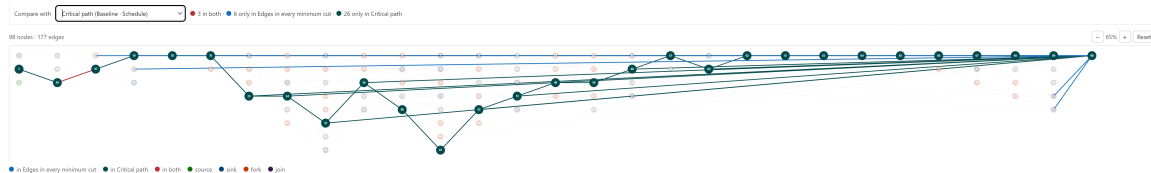
One water network, three questions: EPANET Net3

97 nodes, 119 edges, oriented by its own hydraulic simulation; converted once.

- **Reliability:** worst-served junction 0.81; [0.35, 0.99] under $\pm 5\%$.
- **Capacity:** whole demand met, 680.1 L/s; river pump out, 43.3 L/s (6.4%).
- **Schedule:** 74 days along 28 activities; [59, 89] days under intervals.

For the engineer: the **river pump** (95–97, red) is in every degraded minimum cut *and* on the restoration critical chain. Reinstating it is the step both analyses point to, which neither shows alone.

Corrected since submission: the super-sink edges were missing from the flow graph. Errata sheet provided.



Degraded minimum cut (blue) against the baseline critical path (dark): 95, 97 and the super-sink in both.

Answers, contributions and limits

Answers

- ① Yes, on the model's four assumptions.
- ② Yes, for the probabilistic interpretation.
- ③ Exact for numbers and intervals, sound for p-boxes; cost set by conditioning width.
- ④ Completion time in two runs; floats exact at a cost set by bypass width.
- ⑤ Yes: a registered package and a local no-code interface.

Limits, stated in the thesis

- Practical to conditioning width about 18 (about 50 nodes, moderate reconvergence); p-boxes on small networks.
- Capacity is point-valued.
- Acyclicity is a property of one operating snapshot.
- The interface is demonstrated, not yet studied with engineers in use.

Next: a depth-limited hybrid for larger networks; interval capacities; cyclic networks; a study with engineers in use.

Demo:

`information-propagation-framework.dev`

Thank you

Questions welcome

Backup: how does an engineer get their network in?

- **Structure:** an edge list or an adjacency matrix, the form any DAG description reduces to. The parser detects which, validates acyclicity and builds the graph object once.
- **Inputs:** one JSON file per analysis, which declares its value form (number, interval or p-box), so the same network carries probabilities, capacities and durations side by side.
- **From a domain model:** Net3 came from its EPANET .inp file by a script that orients each pipe by its simulated flow and derives the three input files from pipe diameters, pump curves and published break rates (in the thesis-data repository).
- **Limit, stated in the thesis:** the package itself does not read domain formats such as EPANET or a GIS export; that conversion is a script per source.

Backup: what would you do next?

Each item follows from a limitation stated in the thesis (Chapter 11).

- **Scale:** a depth-limited hybrid, exact to a stated depth and bounded beyond it, for networks past about fifty nodes of moderate reconvergence.
- **Theory:** a soundness proof for the tighter p-box recombination operator, validated sound on every configuration tested but not proved.
- **Schedule:** gate the domination split by its own bypass-set cost in place of a fixed threshold; that threshold sent Net3 to the conservative enclosure.
- **Capacity:** interval and p-box capacities, direct for the flow value by monotonicity, open for the cut structure.
- **Model:** cyclic networks by strongly connected components or unrolling, and an orientation per operating state (Net3's degraded case).
- **Use:** a study of the interface with engineers who do not program.

Backup: what changed since submission?

Chapter 6. “Exactly $2^{|\mathcal{F}|}$ minimum cuts” becomes “at most”: $S^* \cup R$ is a minimum cut iff $R \subseteq \mathcal{F}$ is closed under residual reachability (Picard and Queyranne, 1980). Implementation corrected; every Chapter 6 result re-run and unchanged (Configuration A: 4 cuts; B: 1).

Chapter 10. The flow run used the edge list without the super-sink and its 58 demand edges, and reported 1,837.9 L/s, above the total demand of 680.1. The algorithms were correct; the server in the interface layer ignored the 58 entries without reporting them.

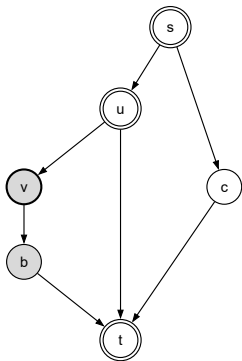
	Submitted	Corrected
Baseline throughput	1,837.9 L/s	680.1 L/s (all demand)
Baseline minimum cut	4 supply edges	the 58 demand edges
Degraded throughput	954.6 L/s	43.3 L/s (6.4%)
Degraded minimum cuts	same 4	24; river pump in every one
Overlay with critical path	“no node in common”	river pump on both

Checked against an independent Edmonds–Karp computation. Reliability and schedule unchanged.

Backup: why does Net3 deliver so little with the river pump out?

- With the river pump out, the sources are the lake pump and one discharging tank.
- In the snapshot orientation they reach only six junctions (7, 8 and 77–81), which are fully served; the other 52 receive less than their demand.
- The derated trunk main does not bind: its only feed is the river pump.
- Physically, the lake pump could push water through pipes that ran the other way at the snapshot. The oriented model cannot represent that. A degraded state needs an orientation of its own.

Backup: how can interval floats be exact when the problem is NP-hard?



Relative to node v : *dominated* durations (every route through them passes v) are fixed at one bound, *incomparable* ones at the other, and only the *bypass* set H_v , which shares routes with v and can route around it, is enumerated.

The exact floats then cost $2 \sum_v 2^{|H_v|}$ propagations in all, against 2^k for k interval durations.

Backup: how are p-boxes combined at a diamond?

Chain and union steps: the quantities combined are independent for the same structural reason the point computation is exact, so p-box arithmetic under independence applies (Williamson and Downs).

The conditioning step is the convex combination

$$W \cdot A + (1 - W) \cdot B$$

where A , B depend on the same upstream reliabilities, and W , $1 - W$ are one variable. Convolving the terms as if independent is **unsound**: probability mass leaves $[0, 1]$, by up to 0.34 on the benchmark grid.

The operator: at each discretisation level of W , weight the branches by w and $1 - w$, combine them under a dependence bound, mix across levels, and envelope the results built on W 's two bounding functions. Taking one conditioning node at a time reproduces the mixture over all $2^{|C|}$ states. The output is clipped to $[0, 1]$, with a guard that raises an error if the excursion is more than round-off.

Backup: is the p-box result sound?

Two dependence bounds for the branches

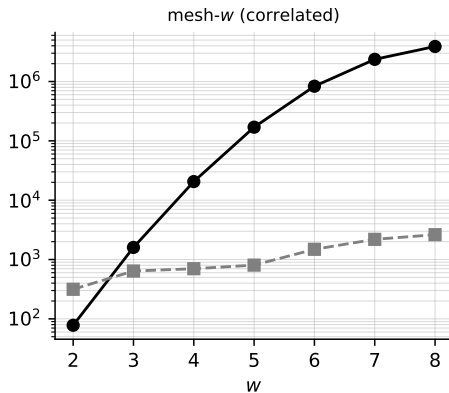
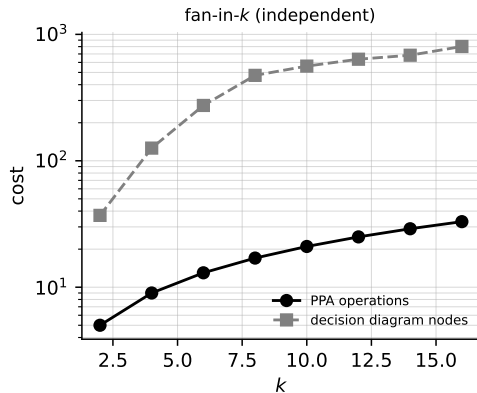
- **Fréchet–Hoeffding**: sound for *any* dependence by a classical theorem; can be wide, near vacuous on hard cases.
- **Positive dependence** (default): A and B both rise with the shared upstream reliabilities. Tighter; no violation in 50 configurations against Monte Carlo. **Not proved**: no theorem covers positive but otherwise unknown dependence, and the ordering-of-dependence route fails (the distributions cross).

Fréchet is used wherever a claim must rest on proof alone.

What it buys

- Band width 0.18 under high reliability and weak reconvergence, 0.70 under deep reconvergence with broad uncertainty; no violations in the sweep.
- For a requirement $x^* = 0.95$: $P(b(v) \leq x^*)$ certified to a band of 0.02–0.12 from one propagation.
- Simulation would need 267 to 9,604 samples, and gives an estimate, not a bound.

Backup: is it faster than a decision diagram?



Sub-problem count against sifted BDD node count on the two adversarial families (fan-in- k , mesh- w).